Q1.

(a) $i = 7\% \Rightarrow \delta = ln1.07 = 0.0676586$. Work in £ millions

$pv(grid) = 20 \times 0.12 \times \bar{a}_{\overline{20}|} = 2.4 \times 10.96062 = £26.305m$ [0.5]

$pv(subsidy) = 40 \times 0.12v + 40 \times 0.12 \times 1.025v^2 + \cdots + 40 \times 0.12 \times 1.025^{19}v^{20}$ [1]

$= 4.8/1.025(1.025v + 1.025^2v^2 + \cdots + 1.025^{20}v^{20})$ [1]

$= 48/1.025a^{j}_{\overline{20}|}$ where $\frac{1}{1+j} = \frac{1.025}{1.07} \Rightarrow j = \frac{1.07}{1.025} - 1 = 4.39024\%$ p.a. [1.5]

$= 48/1.025 \times 13.132536 = £61.498m$

Maximum price is £26.305m+£61.498m=£87.803m [1]

(b) Use real values as of 1 January 2020. Denote real rate of return by *i'* with force of interest δ. Working in £millions

$pv(grid) = \int_0^5 \frac{2.4}{1.025^t} e^{-\delta t} dt = 2.4\bar{a}^{*}_{\overline{5}|}$ where force of interest $\delta^* = \delta - ln1.025$ [2]

$pv(subsidy) = 4.8/1.025a_{\overline{5}|}$ at rate i' [1]

$pv(sale\ price) = 72/(1.025)^5v^5$ at rate i' [1]

So EoV is $82 = 2.4\bar{a}^{*}_{\overline{5}|} + 4.68293a_{\overline{5}|} + 63.6375v^5$

At 5% EoV $= 2.4 \times 4.71052 + 4.68293 * 4.32947 + 63.6375 \times 0.783526 = 81.4415$

At 4% EoV $= 2.4 \times 4.82272 + 4.45182 * 4.32947 + 63.6375 \times 0.821927 = 84.7275$ [1]

By interpolation $i' = 4\% + \frac{84.7275-82}{84.7275-81.4415} \times 1\% = 4.8300\%$ per annum [1]

COMMENTS: This was the most challenging question in the exam and was largely not done well. Many students in (b) recognised that the constant inflation rate allowed them to use the relationship with the money yield, but couldn't follow through on the calculations.

Q2.

(a) Let i_t be the rate of interest in year t and S_{10} be the accumulation of the unit investment after 10 years.

$E(S_{10}) = E[\prod_{k=1}^{10}(1 + i_k)]$ [1]

$= \prod_{k=1}^{10} E[1 + i_k]$ as $\{i_k\}$ are independent [1]

$= (E[1 + i_1])^{10} = (1.06)^{10} = 1.79085$ [1]

(b) $E(S_{10}^2) = E[(\prod_{k=1}^{10}(1 + i_k))^2]$ [1]

$= \prod_{k=1}^{10} E[(1 + i_k)^2]$ as $\{i_k\}$ are independent [1]

$= \prod_{k=1}^{10} E[1 + 2i_k + i_k^2]$

$= (E[1] + 2E[i_1] + E[i_1^2])^{10}$ since identically distributed [1]

$= (1 + 2 \times 0.06 + 0.02 + 0.06^2)^{10} = 3.82597$ since $E[i_1^2] = Var(i_1) + E[i_1]^2$ [1]

Therefore $Var(S_{10}) = E[S_{10}^2] - E[S_{10}]^2 = 3.82597 - 1.79085^2 = 0.61883$ [1]

(c) The mean would be 10 times bigger, but the variance would be 100 times larger. [1]

COMMENTS: this question was generally done reasonable well. Many students lost marks for not justifying steps properly. A number of students assumed a lognormal distribution, which was not in the question.

Q3.

(a) $pv = 1000 \times exp\left[-\int_6^8 (0.01t - 0.04)dt\right] \times exp\left[-\int_4^6 (0.08 - 0.01t)dt\right]$ [2]

$= 1000 \times exp\left(-\left[\frac{0.01t^2}{2} - 0.04t\right]_6^8\right) \times exp\left(-\left[0.08t - \frac{0.01t^2}{2}\right]_4^6\right)$ [1]

$= 1000 \times exp\left[-\left(\frac{0.01(64-36)}{2} - 0.04(8-6)\right)\right] \times exp\left[-\left(0.08 \times 2 - \frac{0.01(36-16)}{2}\right)\right]$ [1]

$= 1000 \times exp(-0.14 + 0.08 - 0.16 + 0.1) = 1000 \times \exp(-0.12)$

$= £886.92$ [1]

(b) $886.92\left(1 + i^{(12)}/12\right)^{12\times 4} = 1{,}000$ [1]

$\Rightarrow 1 + \frac{i^{(12)}}{12} = \left(\frac{1000}{886.92}\right)^{\frac{1}{48}} = 1.0025031$

$\Rightarrow i^{(12)} = 12 \times 0.0025031 = 3.0037\%$ p.a convertible monthly [1]

(c) $av = \int_0^3 100e^{0.03t} \times exp\left(\int_t^3 0.05ds\right) \times exp\left[\int_3^6 (0.08 - 0.01s)ds\right] \times \exp\left[\int_6^{10} (0.01s - 0.04)ds\right] dt$ [2]

$exp\left(\int_t^3 0.05ds\right) = \exp[0.05s]_t^3 = \exp(0.15 - 0.05t)$ [1]

$$exp\left[\int_3^6 (0.08 - 0.01s)ds\right] = exp\left[0.08s - \frac{0.01s^2}{2}\right]_3^6 = \exp\left(0.08 \times 3 - \frac{0.01(36-9)}{2}\right)$$
$$= \exp(0.105)$$

[1]

$$exp\left[\int_6^{10}(0.01s-0.04)ds\right]=exp\left[\frac{0.01s^2}{2}-0.04s\right]_6^{10}$$
$$=\exp\left(\frac{0.01(100-36)}{2}-0.04\times 4\right)=\exp(0.16)$$

[1]

$\Rightarrow av=(100e^{0.15})(e^{0.105}e^{0.16})\int_0^3 e^{0.03t}e^{-0.05t}dt=100e^{0.415}\left[\frac{e^{-0.02t}}{-0.02}\right]_0^3=$
$5000e^{0.415}\,(\mathrm{e}^0-\mathrm{e}^{-0.06})\quad=£440.95$

[1]

COMMENTS: This question was reasonably well done, though a good number of students made (a) harder for themselves by trying to take the ratio of a PV function to time zero rather than going for the integral directly. IN (b), students who did badly usually didn't reognise $i^{(12)}$ was being asked for. In (c), many students were not clear on valuing the payment period and didn't get correct limits on the integrals.

Q4.

(a) Working in £,000

Let X = Nominal amount of ZCB
Y = Nominal amount of FI bond

$V_L=200v^{10}@7\%=101.6699$ [1]
$V_A=101.6699\times 10\%+Xv^{12}+0.05Ya_{\overline{16|}}+1.05Yv^{16}$ [1]

1st condition: $V_A=V_L$
$\Rightarrow 101.6699\ =10.16699\ +0.444012X+0.828004Y$
$\Rightarrow 91.50287=0.444012X+0.828004Y$ [1]

2nd condition: $V_L'=V_A'$
$V_L'=10\times 200v^{10}=1016.6986$ [1]
$V_A'=0\times 10.16699+12Xv^{12}+0.05Y(Ia)_{\overline{16|}}+16\times 1.05Yv^{16}$
$=5.328144X+0.05Y\times\frac{10.107914-16v^{16}}{0.07}+5.690741Y$ [2]

$\Rightarrow 1016.6986=5.328144X+9.039427Y$ [1]

Combining 1st and 2nd condition:

$$\Rightarrow 1016.6986-91.50287\times\left(\frac{5.328144}{0.444012}\right)=\left(11.74615-1.022906\times\left(\frac{5.328144}{0.444012}\right)\right)Y$$

$\Rightarrow 81.33589=0.896618Y$ [1]
$\Rightarrow Y=90.71410$ i.e. Y = £90,714 [1]
$\Rightarrow X=\frac{1016.6986-9.039427*90.71410}{5.328144}=36.91623$ i.e. X = £36,916 [1]

(b) Amount invested in ZCB is $Xv^{12}=36{,}913.23v^{12}=£16{,}391$ [1]

Amount invested in FI bond $=101{,}670*.9-16{,}391=£75{,}111$ [1]

(c) Given the three different assets and the liability is a single point, the spread of the assets is greater than the spread of the liabilities. [1]
Hence, Redington's third condition is satisfied and the company is immunised. [1]

COMMENTS: Although many students made allowance for the bank account, few worked through the full implications. Otherwise, most students made good progress with *V* and *V'* calculations, with most errors suggesting weaknesses in material from the first semester. Part (b) was also first semester material, but a number of students struggled.

Q5.
(a) A replicating portfolio for a contract is a portfolio of assets such that the value of the assets equals the value of the contract at all times t>0. [1]

(b)(i) $K_0 = \left(5 - 0.25e^{-0.03*\frac{2}{12}}\right)e^{0.03\times\frac{6}{12}} = 4.82305$ [1]

(b)(ii) An investor can borrow £5 at the risk-free rate, buy one share at £5 and enter into the forward contract to sell the share in six months time for £4.85. [1]

Initial cashflow is zero [0.5]

After two months the 25p dividend can be invested at the risk free rate.

After 6 months share can be sold for £4.85, the dividend proceeds are worth $0.25e^{0.03\times\frac{4}{12}}$ and the borrowing repaid at $5e^{0.03\times\frac{6}{12}}$.

Net cashflow $4.85 + .025e^{0.01} - 5e^{0.015} = 0.026948$ [2]

Therefore the investor has made a risk-free future profit [0.5]

COMMENTS: This question was generally well done, although some struggled with basic knowledge. Several loss marks for either constructing the arbitrage the wrong way round or not explaining it fully.

Q6.
(a) Let y_n be the spot yield for term *n*

$105(1 + y_1) = 105 + 6 = 111 \Rightarrow y_1 = 5.7140\%$ [1]

$104 = 6(1 + y_1)^{-1} + 111(1 + y_2)^{-2} \Rightarrow y_2 = \left(\frac{111}{104-6(1.057140)^{-1}}\right)^{0.5} - 1 = 6.2510\%$ [1]

$103 = 6(1 + y_1)^{-1} + 6(1 + y_2)^{-2} + 111(1 + y_3)^{-3}$

$\Rightarrow y_3 = \left(\frac{111}{103-6(1.057140)^{-1}-6(1.062510)^{-2}}\right)^{1/3} - 1 = 6.4540\%$ [1]

(b) $f_{0,1}, f_{0,2}$ and $f_{0,3}$ are the same as the spot rates y_1, y_2 and y_3 respectively [1]

$f_{1,2} = \frac{(1+y_2)^2}{1+y_1} - 1 = 6.7894\%$ [1]

$f_{2,3} = \frac{(1+y_3)^3}{(1+y_2)^2} - 1 = 6.8632\%$ [1]

$f_{1,3} = \sqrt{\frac{(1+y_3)^3}{1+y_1}} - 1 = 6.8263\%$ [1]

COMMENTS: Part (a) was not done very well, with many students not using the bonds as described in the question. Part (b) was generally well done.